

# Course 6

## METHODOLOGY FOR DEVELOPING A DB

DATABASES

# CONTENTS

- Organizing a database
- Objectives for developing a DB
- Steps in developing a DB
- DB efficiency

# Organizing a database

- Organizing data inputs
- Organizing data storing
- Organizing data protection
- Organizing database connections with other applications

# Organizing data inputs

- source of data (primary documents (receipts, goods received note GRN - NIR), files (from editors, from programming languages) etc.)
- data updating (type (I, U, D), frequency (daily, monthly, etc.), reorganizing (in case of incidents, periodically) etc.);
- data input control (type (character, numeric, etc.), content (values, states), format (binary, unpatched), frequency (daily, monthly, etc.), etc.).

# Organizing data storing

- principles of storage:
  - internal/external memory – memory allocation,
  - access ,
  - storage type - depending on the data model,
  - links between data (type, no).
- storage optimization:
  - minimum and controlled redundancy,
  - optimal space,
  - allocation schemas,
  - encrypting techniques,
  - reorganization strategies

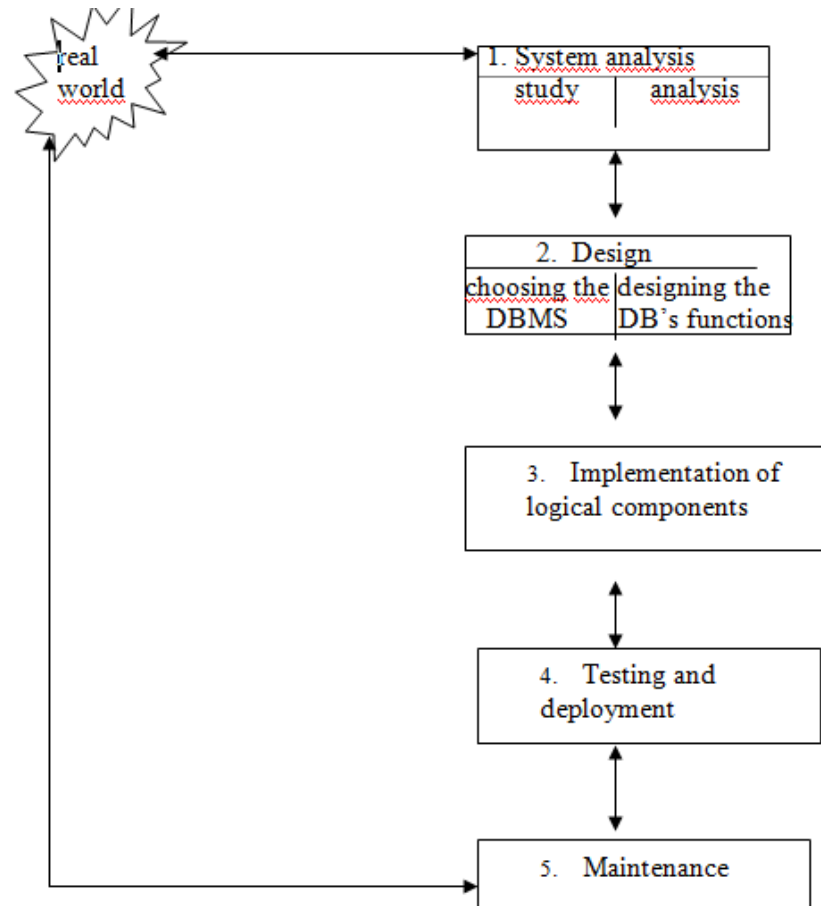
# Organizing data protection

- tools provided by the DBMS:
  - authorizing access,
  - log files,
  - archiving.
- personal methods:
  - passwords,
  - special routines,
  - antivirus.

## **Objectives for developing a DB**

1. Partitioning: the same data must be used in different ways by different users.
2. Openness: data must be easily adaptable to changes (updating the structure, new data types, etc.).
3. Efficiency: data storage and processing should be at low cost (time and space), in order to make it worth to obtain the result.
4. Reuse: existing processed data can be reused in different applications.
5. Retrieval: requests must be easily addressed by all users, according to different criteria.
6. Access: data must be located in various ways in order to be quickly and easily accessed.
7. Modularization: DB development should be modularly made and there should be the possibility of teamwork.
8. Protection: ensure data security and integrity.
9. Redundancy: minimum and controlled redundancy is provided by the implementation of a data model and by using design techniques for the database.
10. Data independence to programs from logical and physical levels.

# Steps in developing a DB



# 1. System analysis

The purpose of this activity is to highlight the application requirements and resources (study) and to evaluate these requirements by modeling (analysis).

- STUDYING the existing situation is achieved by:
  - defining general characteristics of the organization;
  - identifying the carried activities;
  - identifying the existing resources (informational, human, technical equipment, financial, etc.);
  - identifying the needs for processing.
- The ANALYSIS of the existing system follows the investigation (study) and uses the information obtained from it.

The analysis is a modeling (conceptual) activity and is performed under three aspects: **structural, dynamic, functional**.

# The structural analysis

- *The structural analysis* highlights the data structures and the links between them. The **entity-relationship** is the used technique and it is described below.
  - 1) Identifying the entities: events, processes, actual or abstract objects (nouns in the described activity) - e.g. Person, Products, Customers.
  - 2) Identifying the relationships between entities as significant links of a particular type (verbs in the described activity) – e.g. has a, is made of, etc.
  - 3) Identifying the attributes that characterize each entity - e.g. Brand, Name, Address.
  - 4) Setting attributes that uniquely identify the entity realizations as keys.

The result: the static (structural) model (entity-relationship diagram). The *Entity-Relationship technique* was introduced by P.P.S. Chen in 1976.

The E-R diagram can be generated by CASE (Computer-aided software engineering) software tools (e.g. Oracle SQLDeveloper, JDeveloper).

# The dynamic analysis

*The dynamic analysis* highlights the behavior of the system elements that are faced to certain events. One of the techniques that is used is the state-transition diagram and it is described above.

- 1) Identifying the states in which the system components can be found.
- 2) Identifying the events that can make a component pass from one state to another.
- 3) Establishing the allowed transitions between states.
- 4) Building the diagram.

The result: the dynamic model (state-transition diagram).

E.g. the entity ORDER

-> state: issued, received, pending, processed, canceled, etc.

-> event: placing an order, receiving an order, processing an order, product is out of stock, etc.

-> allowed transition: pending-> processed, not the other way!

<https://www.smartdraw.com/state-diagram/>

# The functional analysis

*The functional analysis* highlights how informational requirements of the system (flow processing) can be met, by which **inputs** are transformed into **outputs**. The technique that is used is the data flow diagram. According to this technique we have the following steps:

- 1) Delimiting the coverage of the system;
- 2) Identifying the data sources;
- 3) Identifying the pattern of movement and processing of data;
- 4) Identifying the obtained results.

The result: the functional model (data flow diagram).

## 2. Design

- Starting from the models made in the analysis activity, the database structure can be designed, following two subtasks: selecting the DBMS and designing DB's features.
- *a)* when *choosing the DBMS* we have to consider two aspects: the application (user) requirements and the technical performances of the DBMS.
- *b)* *designing the DB's functions* is achieved by: designing DB's schemas and designing specialized functional modules.

## The application requirements

- amount of data expected to be stored and processed in the database;
- complexity of the problem to be solved;
- frequency of input / output operations;
- conditions regarding data protection;
- necessary database operations (loading / validation, updating, retrieval, etc.);
- specific features of the activity for which the database is created.

## The technical performances of a DBMS

- data model that is implemented;
- trend of using the DBMS in the market;
- minimum required hardware configuration;
- DBMS's programming languages;
- usage facilities provided for different categories of users;
- DBMS's limitations;
- optimizations performed by the DBMS;
- technical facilities:
  - working with distributed environment and data concurrency;
  - multimedia elements;
  - CASE elements;
  - communication interfaces;
  - self-documenting;
  - specific tools that are offered.

# Conceptual schema

- Conceptual schema's design starts with identifying the data set required by the system. The data is then integrated and structured in a schema (e.g. for RDB the most used technique is normalization). The next steps have to be followed.
- Setting the initial conceptual schema that is derived from the entity-relationship model (structural analysis). Each entity in the model is transformed into a collection of data (files), and the keys are defined for each relationship. If there are isolated collections, they will link to other collections by keys, relationships being resulted (1: 1, 1: M, M: N).
- Gradual improvement of the conceptual schema by removing anomalies (e.g. the five NF).
- Establishing the final conceptual schema should provide a balance between exploitation performances and updating requests (e.g. a higher NF provides update performances but gives a higher response time).

# External schema

External schema's design is meant to specify each user's view over the database, by following these steps:

- The data required for each view is identified from the conceptual schema.
- The data is structured logically in sub-schemas given the ease of use and user requirements.
- External schema becomes operational by building views with the DBMS and by granting access rights. Data in a view may come from one or more collections and does not occupy physical space.

# Internal schema

- Internal schema's design involves establishing the structures for physical storage of data and defining ways to access the data. They are specific either to the DBMS (allocation schemas), or to the OS. Internal schema's design contains the following operations:
  - estimating the physical space for the database and defining a physical model for allocation (if the DBMS allows)
  - defining indexes for direct access by keys;
  - building clusters that mean physical grouping of data from the database in order to allow faster access to certain users. The data may come from one or more collections.

## Designing the functional modules

- informational flow;
- loading and manipulation modules of data;
- specialized interfaces;
- integration of the designed elements with DB organization and functioning.

### 3. Implementation of logical components

- The logical components of a database are the application programs that are mostly developed in the chosen DBMS.
- The programs are implemented according to the functional modules that were designed in the previous step.
- Logical components take into account the outputs, inputs, processing and data collections.
- Technical, operational and presentation documentation is developed in parallel with the implementation of application programs.

## 4. Testing and deployment

- The DB functions are tested first with test data, and then with real data.
- Loading data in the database and manipulation procedures are performed by the beneficiary with the assistance of the designer.
- Application documentation is finalized.
- The beneficiary uses the DB with the documentation.

## 5. Maintenance

- Immediately after deploying the DB, and after that continuously, there may be disrupting factors that produce changes in the database.
- The changes need to be taken into account easily and "on the fly" by the DB.
- The factors are: organizational, due to technical progress, resulting from new requirements of the beneficiary, changed methodologies or laws, etc.

# Steps in developing a DB

- 1. System analysis
  - study
  - analysis
    - structural analysis => entity-relationship diagram
    - dynamic analysis => state-transition diagram
    - functional analysis => data flow diagram
- 2. Design
  - choosing the DBMS
    - application requirements
    - technical performances
  - designing the DB's functions
    - designing the DB schemas (conceptual, external, internal)
    - designing the functional modules
- 3. Implementation of logical components
- 4. Testing and deployment
- 5. Maintenance

*The steps* that need to be followed in order to determine the DB efficiency:

1. Study the initial cost for implementation:

- cost of necessary software;
- cost of required hardware;
- cost imposed by the transition from the existing system to the DBS;
- cost of necessary resources (human, material, informational etc.);
- cost of technical advice.

2. Assessing the costs of DB functioning:

- cost of deployment;
- cost of testing;
- maintenance costs.

3. Profitability estimations:

- using specific methods (e.g. Cost Benefit Analysis)
- using specialized software and hardware tools for the DB (increase performance, reduce portability).

# Optional homework

1. Choose a DBMS to present its new features vs Oracle

2. Register here by choosing a different DBMS:

[https://docs.google.com/spreadsheets/d/15jSv9NPejf7ogHk2ltmVOz7sxc-Jf9UvS\\_d9RNeptb4/edit?gid=0#gid=0](https://docs.google.com/spreadsheets/d/15jSv9NPejf7ogHk2ltmVOz7sxc-Jf9UvS_d9RNeptb4/edit?gid=0#gid=0)

3. Upload the homework on the online platform

# Homework

- The document must include:
  1. The type of the DBMS (relational, object-oriented, network, hierarchical)
  2. Comparison with Oracle DB
  3. The query language (2-3 examples)
  4. Info about the latest release (version + new capabilities)